Bank Exploration Under Different Market Conditions

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1 Setup

Borrowers

Borrowers live in two periods. In each one, they could request a loan, for which they are charged R_0 and R_1. Borrowers always demand loans in the first period, but there is a chance of ρ of the asking for a second loan in the next period. If two loans are offered, they choose the one with the lower rate. Their utility in each period is described as:

$$U_i = 1 - \delta R_i$$

. Each borrower has a probability of paying the loan of θ in the first period.

Banks

Banks decide to set the interest rates R_0 and R_1. After observing the payment history of the first period, banks can update their beliefs about the borrower they observe by a Bayesian update. The probability of repayment updates to θ_g under a repayment, $x = 1$, and updates to θ_b under a non-repayment, $x = 0$. The following holds between these probabilities:

$$\theta = \theta\theta_g + (1 - \theta)\theta_b$$

Considering this, the bank has to optimize a profit function in two periods by setting the interest rate it will charge a borrower with a given profile θ.

The profit for the bank when lending to a borrower of type θ is, when the information of repayment is available is the following:

$$\begin{aligned} \pi(\theta) = \quad & p_0(\theta, R_0, \sigma_{-i})(\theta R_0 - 1) + \rho[\theta(p_1(\theta, R_1(x = 1), x = 1, \sigma_{-i})\theta_g R_1(x = 1) - 1) \\ & + (1 - \theta)(p_1(\theta, R_1(x = 0), x = 0, \sigma_{-i})\theta_b R_1(x = 0) - 1)] \end{aligned}$$

In the case the repayment information won't become available, the bank can only set the strategy for θ, and not for the repayment realizations:

$$\begin{aligned}\pi(\theta) = \quad & p_0(\theta, R, \sigma_{-i})(\theta R_0 - 1) + \rho[\theta(p_1(\theta, R_1, x=1, \sigma_{-i})\theta_g R_1 - 1) \\ & \qquad + (1-\theta)(p_1(\theta, R_1, x=0, \sigma_{-i})\theta_b R_1 - 1)]\end{aligned}$$

σ_{-i} represents the strategies for the other banks, which affects the probability of granting the loan.

2 Monopolistic Case

We begin by analyzing the case of a monopolistic lender. Under monopolistic conditions, the lender can set:

$$R^M = \frac{1}{\delta}$$

Thus, the monopolist extracts all surplus from the borrower. The monopolist will lend until $\pi(\theta) = 0$. The benefit of exploring is determined by:

$$\psi^M(\theta) = \frac{\theta_g}{\delta} - 1$$

The bank will explore riskier profiles, such that it would reach a zero profit condition in the margin:

$$\pi(\theta^M) = \theta^M R^M - 1 + \rho\theta\psi^M(\theta) = 0$$

$\theta^M < \delta$, so the firm explores in the first period since it loses and gets a return in the future.

3 Competition with Public Information

We now analyze the case of competition between lenders when information is public. Under competitive conditions with public information, we have the Bertrand result:

$$R^C(\theta) = \frac{1}{\theta}$$

When information is public, competition drives interest rates down to the point where lenders break even. The benefit of exploring (acquiring information) is zero because the information is public and available to all lenders.

4 Competition with Private Information

Pure Strategy Equilibria

When information is private, the dynamics become more complex. In the second period, interest rates do not decrease as much as in the public information case. This is because the information is not shared, creating information asymmetry between lenders. The equilibrium is different even for borrowers with profitable risk profiles.

Since the non-explorer does not have access to private information, there are profitable deviations from pure equilibrium strategies that make this unfeasible as a Nash Equilibrium:

1. Given that the explorer sets rates for good borrowers at a lower level than the non-explorer, if the non-explorer sets $R = \frac{1}{\theta}$, it will attract only poor risk profiles.
2. If the non-explorer sets $R = \frac{1}{\theta_b}$, the explorer bank has incentives to increase rates for good borrowers.
3. If the explorer bank sets an interest rate higher than $1/\theta$, the non-explorer bank makes a profit by undercutting the explorer bank and capturing all the market.

Mixed Strategy Equilibria

Due to the strategic interactions between explorer and non-explorer banks, pure strategy equilibria may not exist. We now characterize the mixed strategy equilibria. These will be summarized by a distribution of rates for a given profile θ for the explorer $F^E(.)$ and the non-explorer bank $F^{NE}(.)$:

Profit for Explorer

The following conditions must hold for the explorer bank:

$$[1 - F^{NE}(R)](\theta_g R - 1) = [1 - F^{NE}(R')](\theta_g R' - 1)$$

This is because the explorer bank should be indifferent between any strategy for $R \in [1/\theta, 1/\theta_g]$. We know that the non-explorer bank should charge more than $1/\theta$. Considering that, the benefit of setting $R = 1/\theta$ is $\frac{\theta_g}{\theta} - 1$. Using this and the indifference condition, we can infer F^{NE} in the equilibrium:

$$1 - F^{NE}(R) = \frac{\theta_g - \theta}{\theta(\theta_g R - 1)}$$

$$F^{NE}(R) = \frac{\theta\theta_g R - \theta_g}{\theta\theta_g R - \theta}$$

This when $R < 1/\theta_b$. For any $R \geq 1/\theta_b$ we get $F^{NE}(R) = 1$.

Profit for Non-Explorer

For the non-explorer, the profit from lending is:

$$\theta[1 - F^E(R)](\theta_g R - 1) + (1 - \theta)(\theta_b R - 1)$$

Similarly to the explorer, this profit should be equal across R. If the non-explorer chooses $R = \frac{1}{\theta_b}$, the profit is zero. Using this condition, and the indifference across options, we can infer F^E in equilibrium:

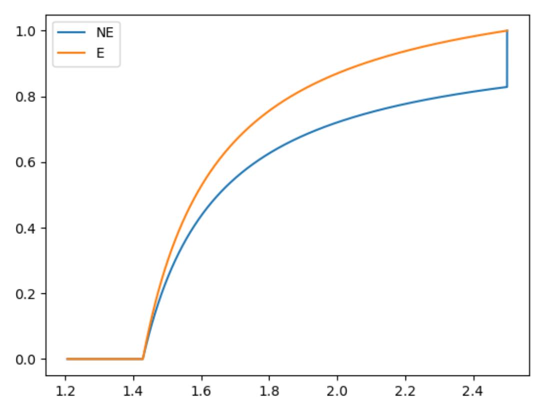
$$\begin{aligned} 1 - F^E(R) &= -\frac{(1-\theta)(\theta_b R - 1)}{\theta(\theta_g R - 1)} \\ F^E(R) &= 1 + \frac{(1-\theta)}{\theta} \times \frac{(\theta_b R - 1)}{(\theta_g R - 1)} \end{aligned}$$

This is as long $R > 1/\theta$. For $R < 1/\theta$, $F^E(R) = 0$.

From Figure 1, we can observe that the interest rate charged by the non-explorer bank is higher (the distribution of NE stocastically dominates the distribution of E).

Comparison of strategies

Figure 1: Comparison of the CDF of R for explorer and non-explorer banks



$\theta = 0.7$, $\theta_b = 0.4$, $\theta_g = 0.82$

Exploration within this framework

Given this solution, the benefit of exploring is positive and equal to $\frac{\theta_g}{\theta} - 1$. In the first period, banks would compete for this benefit.

Even if negative profits occur initially, banks will lend to firms if they expect to earn $\frac{\theta_g}{\theta} - 1$ with probability θ:

$$\theta R_0 - 1 + \rho(\theta_g - \theta) = 0$$
$$R_0 = \frac{1 - \rho(\theta_g - \theta)}{\theta}$$

R_0 will be lower under a high continuation value, and if the signal updates the probability significantly.

It is still pending to check the benefit of borrower under this setups.

5 Borrower welfare

In the monopolist case, the welfare is zero. In the competence with public information, the welfare is $1 - \delta/\theta$ in the first period, and $1 - \delta/\theta_g$ and $1 - \delta/\theta_b$ in the second period according to the pay outcome.

The private information case is quite complex. The utility in the second period for the bad-borrower ex-ante is given by the utility of taking the lottery from the non-explorer bank:

$$u_1(x = 0) = 1 - \delta \int_{1/\theta}^{1/\theta_b} R dF^{NE}(R)$$

And in the case of the good-borrower, the utility is given by having the lower rate between the good and the bad borrower, which is described by:

$$u_1(x = 1) = 1 - \delta \int_{1/\theta}^{1/\theta_b} R d(F^{NE}(R) + F^E(R) - F^{NE}(R) \times F^E(R))$$

We aggregate the utilities across periods, considering the following:

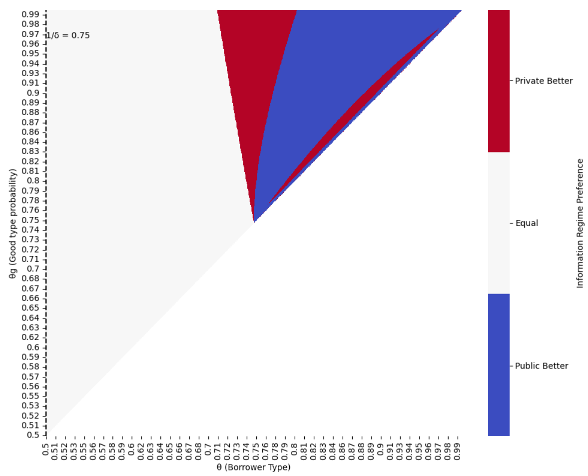
$$u = u_0 + \rho(\theta u_1(x = 1) + (1 - \theta)u_1(x = 0))$$

Also, it is relevant to notice that the maximum interest rate the bank can charge is $1/\delta$. This feature of the models gives us a boundary for the lending decisions, if the interest rate that the bank would charge is higher than this limit, there is no lending, and in consequence, no exploration.

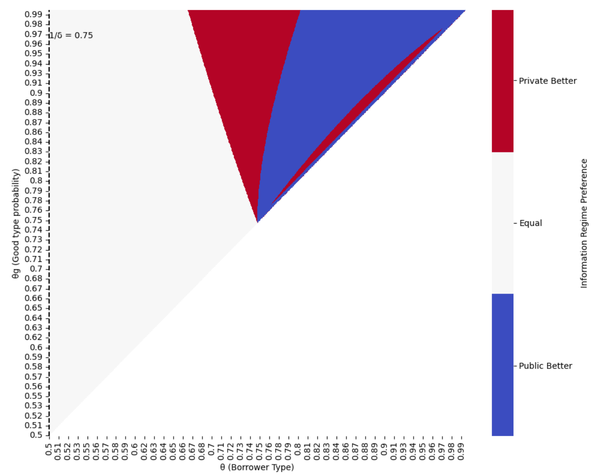
In the monopoly case, this matter doesn't change anything in the analysis, since the bank already charges the maximum interest rate. However, in the competition with public information, the banks won't give credit to borrowers with profiles $\theta < \delta$, since they would incur losses that they wouldn't recover. In the case of the competition with private information, the bank has more incentives to borrow, since it can make profits from the information it can gather from the exploration.

As we can observe in Figure 2c and 3c, the private information setup is preferred by borrowers if θ is close to δ. This comes from the idea that for $\theta < \delta$, the bank doesn't explore, therefore, the borrower is better off under private information since the bank does explore under that setup. Also, the interest rate is lower in the first period, given that exploration is beneficial. The range of θ for which the bank explores is wider if the continuation value ρ is larger. The exploration is narrow if δ is larger.

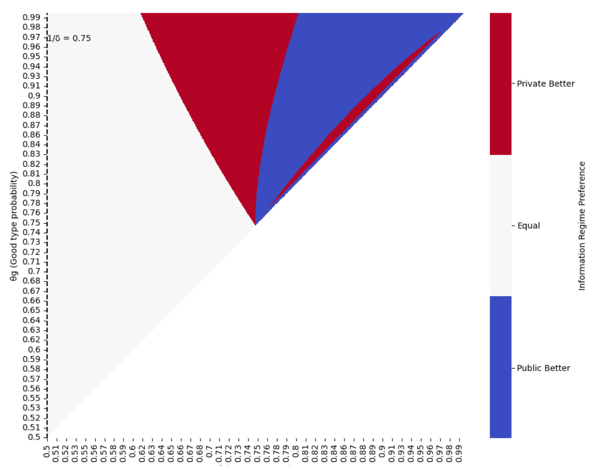
Figure 2: Preferred option for borrower, ρ varying



(a) $\rho = 0.25$

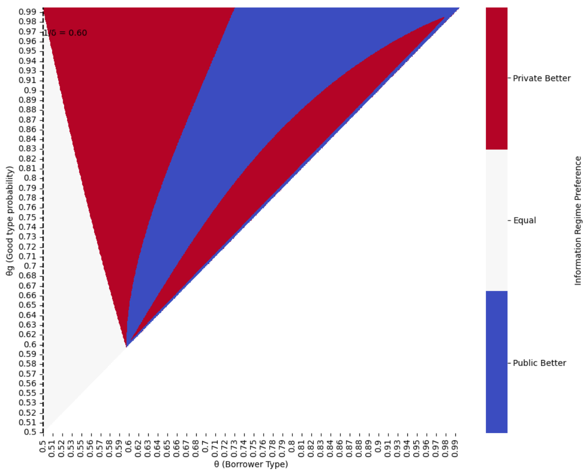


(b) $\rho = 0.50$

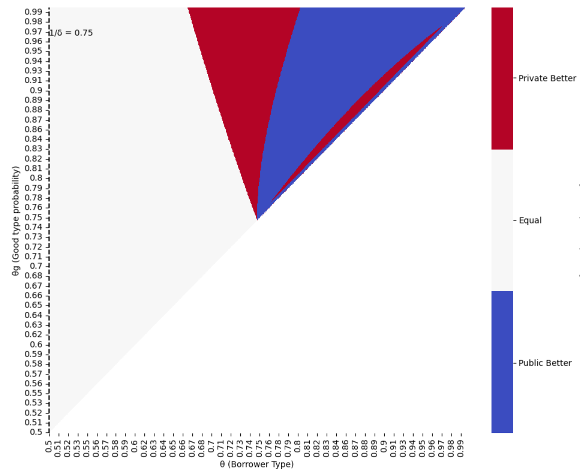


(c) $\rho = 0.90$

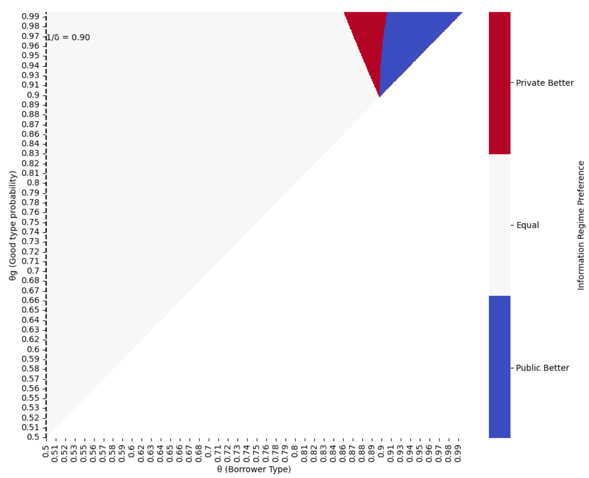
Figure 3: Preferred option for borrower, δ varying



(a) $\delta = 0.60$



(b) $\delta = 0.75$



(c) $\delta = 0.90$

6 Conclusion

This paper makes three primary contributions to our understanding of bank lending behavior under different market conditions:

- We demonstrate that information acquisition through borrower exploration creates tangible economic value, with the benefit of exploration precisely quantified as $\frac{\theta_g}{\theta} - 1$ in competitive markets with private information.
- Second, our analysis reveals that market structure fundamentally alters lending strategies, with monopolistic lenders able to extract full surplus while competitive markets with public information drive interest rates to the break-even point.
- Third, we establish that information asymmetry in competitive markets generates mixed

strategy equilibria where explorer banks generally charge lower interest rates than non-explorer banks, explaining observed interest rate dispersion among seemingly identical borrowers.

These findings have significant implications for understanding credit rationing, relationship banking, and the dynamics of borrower-lender interactions over time. The model provides a theoretical foundation for why banks might initially lend to riskier borrowers despite short-term losses, highlighting the strategic value of information acquisition in dynamic lending relationships. Also, this theoretical model explains why we could observe more dispersion of interest rates for a borrower with the same risk profile.